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Relation:

Relation indicates a relationship between two objects.

- student - teacher relationship.
- Employee and his salary relationship, etc.

Relation used in Mathematics such as less than, "parallel to", "is a subset" "is similar to", etc.

An ordered pair (a, b) is a listing of objects 'a' and 'b' in definite order with 'a' appearing first and 'b' appears in second place. An ordered pair is merely a sequence of length 2.

The most direct way to express a relationship between elements of two sets is to use ordered pairs made up of two related elements. Let $A = \{$ Lucknow, Bhopal, Chennai $\}$ and $B = \{$ Uttar Pradesh, Madhya Pradesh, Tamil Naidu $\}$

There is a relationship " is capital of" between the elements of sets A and B.

Cartesian Product of two sets:

$A \times B = \{(a, b) : a \in A \& b \in B\}$ is called cartesian product of sets A and B. It is an ordered pair.

Let $A = \{1, 3, 5, 9\}$, $B = \{0, 2, 4, 8\}$

then $A \times B = \{(1,0), (1,2), (1,4), (1,8), (3,0), (3,2), (3,4), (3,8), (5,0), (5,2), (5,4), (5,8), (9,0), (9,2), (9,4), (9,8)\}$

Suppose R is less than is a relation between $A \times B$, then

$A R B = \{(1,2), (1,4), (1,8), (3,4), (3,8), (5,8)\}$

It is evident that $A R B$ is a subset of $A \times B$.

We write $A \times A = A^2$,

and $\underbrace{A \times A \times \cdots \times A}_{n\text{-times}} = A^n$.

Total number of distinct elements in $A \times B$.

If $n(A) = m$ and $n(B) = n$, then $n(A \times B) = mn$

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Q: $N \to$ set of all natural nos

Def^n: Let A and B be two sets. A binary relation or relation from A to B is a subset of $A \times B$ and is denoted by R.

i.e.

R is a relation from A to $B \Leftrightarrow R \subseteq A \times B$. Also

$R = \{(a,b) : a \in A,\ b \in B\}$.

$\Rightarrow (a,b) \in R$ then we say that "a is R-related to b" and written as aRb.

If $(a,b) \notin R$ then "a is a not R-related to b" written as $a \not R b$ or $a \bar{R} b$.

Total no. of possible elements in $A \times B$ if $n(A) = m$, $n(B) = n$ will be $= 2^{mn}$.

Domain and Range of a Relation:

The domain of a relation R is the set of all first elements of the ordered pairs which belonging to R and range of R is the set of second elements.

Domain of R = set of first elements of the ordered pairs which belong to R.

$Dom(R) = \{a : a \in A$ and $(a,b) \in R$ for some $b \in B\}$

Range of R = Set of second elements of the ordered pairs which belong to R

$Ran(R) = \{b : b \in B$ and $(a,b) \in R$ for some $a \in A\}$

$Dom(R) \subseteq A$ and $Ran(R) \subseteq B$

Ex1: Let $A = \{1,2,3,4\}$ and $B = \{3,4,5,6\}$, then find a relation (i) $a < b$, such that $(a,b) \in R$ (ii) a and b both are both numbers.

(i) aRb = "a is less than b"

$= \{(1,3), (1,4), (1,5), (1,6), (2,3), (2,4), (2,5), (2,6), (3,4), (3,5), (3,6), (4,5), (4,6)\}$

$Dom(R) = \{1,2,3,4\}$.

$Ran(R) = \{3,4,5,6\}$.

(ii) aRb = "a and b both are odd nos".

$= \{(1,3), (1,5), (3,3), (3,5)\}$

$Dom(R) = \{1,3\}$, $Ran(R) = \{3,5\}$.

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Binary Relation in a set:

Suppose A is a set. A relation R in a set A is the subset of $A \times A$. Clearly, here both the coordinates of the ordered pairs in R are the element of the set. This Relation in the set A is called a binary relation.

Total no. of distinct binary relation from a set A to A if $n(A) = n$ will be 2^{n^2}.

Some operations on sets:

If R and S are two relations, then $R \cap S$ → Intersection of R and S i.e

$$a(R \cap S)_b = aR_b \cap aS_b$$

Also $R \cup S$ → Union of R and S i.e.

$$a(R \cup S)_b = aR_b \cup aS_b.$$

and $$a^{(R-S)}_b = aR_b - aS_b.$$

~~and~~ i.e $aR'_b = a\not{R}y$.

~~For~~ example: Let $R = \{(x,a), (x,b), (y,c)\}$

and $S = \{(x,b), (y,c)\}$.

$R \cup S = \{(x,a), (x,b), (y,c)\}$.

$R \cap S = \{(x,b), (y,c)\}$

$R - S = \{(x,a)\}$.

Types of Relation:

I. Identity Relation: Let A be any set. An important relation on A is that of equality i.e: $\{(a,a) : a \in A\}$, usually denoted by "=". This relation is also called the identity relation or diagonal relation on and is denoted by Δ_A or I_A

i.e $I_A = \{(a,a) : a \in A\}$.

e.g: $A = \{a,b,c\}$ then

$I_A = \{(a,a), (b,b), (c,c)\}$.

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II: Inverse Relation : If R is a relation from set A to B, then the inverse of R is a relation from set B to A and is denoted by R^{-1}:

ie If $R_{A\to B} = \{(a,b) : a \in A \text{ and } b \in B\}$

then $BR^{-1}_A = \{(b,a) : b \in B \text{ and } a \in A\}$

For example: Suppose $R = \{(1,7), (2,3), (3,8), (4,5)\}$.

Then $R^{-1} = \{(7,1), (3,2), (8,3), (5,4)\}$.

Clearly $(R^{-1})^{-1} = R$. Moreover if R is a relation on A, then R^{-1} is also a relation on A.

III n-ary Relation :

Let $\{A_1, A_2 - - - A_n\}$ be a finite collection of sets. A subset R of $A_1 \times A_2 \times - - \times A_n$ is called an n-ary relation on $A_1 \cdots A_n$.

(i) If $R = \phi$, then R is called void or empty relation.

(ii) If $R = A_1 \times A_2 \times \cdots \times A_n$, then R is called the universal relation.

(iii) For $n=1$, R is called unary relation

For $n=2$, R is called binary relation.

For $n=3$, R is called ternary relation.

IV: Reflexive Relation :-

A relation R on a set A is reflexive if aRa for every $a \in A$.

ie every element of A is R-related to itself.

i.e. $(a,a) \in R$, $\forall a \in A$.

ie aRa, $\forall a \in A$.

For example: Let $A = \{1,2,3\}$, then the relation R defined on A as $R = \{(1,1), (1,3), (2,1), (2,2), (3,2), (3,3)\}$

Then R is Reflexive relation since every $a \in A$, $(a,a) \in R$.

Set inclusion $\subseteq$ on a collection of sets is a reflexive relation since every set is a subset of itself.

Relation $R = \{(a,b) \in R^{\oplus} : a \leq b\}$ is a reflexive relation since $x \leq x$ for every $x \in R$.

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(8) v. Irreflexive Relation : A relation R on a set A is irreflexive if for every $a \in A$, $(a,a) \notin R$. i.e., there is no $a \in A$ such that aRa.

For example: Relation $R = \{(x,y) \in R^2 : x < y\}$ is an irreflexive relation since $x < x$ for no $x \in R$.

vi: Non-Reflexive Relation

A relation R on a set A is non-reflexive if R is neither reflexive nor irreflexive, i.e. if aRa is true for some a and false for others.

for example, $R = \{(1,2), (1,1), (2,1), (2,3), (3,3)\}$ on $A = \{1,2,3\}$ is non-reflexive relation.

vii Symmetric Relation :

A relation R on a set A is symmetric if whenever $(a,b) \in R$ then $(b,a) \in R$ i.e. $(a,b) \in R \Rightarrow bRa$.

for example $R = \{(1,1), (1,2), (2,1), (2,3), (3,2)\}$ on $A = \{1,2,3\}$. is a symmetric relation.

viii Asymmetric Relation : A relation R on a set A is asymmetric if whenever $(a,b) \in R$ then $(b,a) \notin R$, i.e. for $a \neq b$ if $aRb \Rightarrow bRa$.

For Example: The relation $R = \{(1,2), (1,1), (2,3), (3,1)\}$ on $A = \{1,2,3\}$.

ix: Antisymmetric Relation :

A relation R on a set A is anti symmetric if whenever aRb and bRa then $a = b$ $\forall a, b \in A$.

for Example, $R = \{(x,y) \in R^2 : x \leq y\}$.

is anti symmetric relation since $x \leq y$ and $y \leq x \Rightarrow x = y$.

Remark : Antisymmetric is not the same as not symmetric. A relation may be symmetric as well as anti-symmetric at the same time e.g., the relation $R = \{(1,3), (3,1), (2,3)\}$ is neither symmetric nor anti symmetric on the other hand $S = \{(1,1), (2,2)\}$ is both symmetric and anti symmetric.

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IX

Transitive Relation:

A relation R on a set A is transitive if whenever aRb, and bRc then aRc i.e. if whenever $(a,b) \in R, (b,c) \in R \Rightarrow (a,c) \in R$.

for example: The relation " is parallel to" on the set of lines in a plane is transitive.

— So the relation < is 'less than' and 'is greater than' are transitive relation on the set of real nos.

Important Properties

1. Reflexivity : $(a,a) \in R$ i.e $aRa, \forall a \in A$
2. Irreflexivity $(a,a) \notin R$, i.e $a \not R a, \forall a \in A$
3. Symmetric/try $(a,b) \in R \Rightarrow (b,a) \in R \ \forall a,b \in A$
4. Asymmetry $(a,b) \in R \Rightarrow (b,a) \notin R \ \forall a,b \in A$
5. Anti Symmetry, $(a,b) \in R$, and $(b,a) \in R \Rightarrow a=b, \forall a,b \in A$
6. Transitivity $(a,b) \in R, (b,c) \in R \Rightarrow (a,c) \in R \ \forall a,b,c \in A$

Examples (1) Let $A = \mathbb{Z}$ the set of integers and let $R = \{(a,b) \in A \times A : a < b\}$ so that R is Relation 'less than'. Is R symmetric, asymmetric or anti symmetric.?

Sol^n: Symmetry: If $a<b$, then it is not true that $b<a$.

So, R is not symmetric.

Asymmetry: Asymmetry. If $a<b$, then $b \not< a$ (b is not less than a)

So R is Asymmetric.

Anti symmetric. If $a \neq b$, then either $a \not< b$ or $b \not< a$.

i.e. if $a<b$ and $b<a$ then $a=b$.

So, R is antisymmetric.

Q Examples of a relation which is

(i) reflexive and transitive but not symmetric
(ii) symmetric and transitive but not reflexive
(iii) reflexive and symmetric but not transitive
(iv) reflexive and transitive but neither symmetric nor

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Soln: Let the set $A = \{1, 2, 3\}$, then the relation

(i) $R = \{(1,1), (2,2), (3,3), (1,3), (1,2)\}$ is reflexive and transitive but not symmetric $(3,1), (2,1) \notin R$.

(ii) $R = \{(1,1), (2,2), (1,3), (3,1)\}$ is symmetric and transitive not reflexive $\because (3,3) \notin R$.

(iii) $R = \{(1,1), (2,2), (3,3), (1,2), (2,1), (3,2)\}$ is reflexive and symmetric but not transitive $\because (1,3) \notin R$.

(iv) Let Z^* be the set of all non-zero integers and R be the relation on Z^* such as $(a,b) \in R$ if a is the factor of b, i.e. $a|b$.

Since $a|a \ \forall a \in Z^*$, hence R is reflexive

If $a|b$ and $b|c$ then $a|c$ i.e. it is transitive.

But if $a|b$ then $b|a$ is not true i.e. $2|16$ but $16|2$ is not true, so R is not symmetric

Again if $6|-6$ and $-6|6$ then $6 \neq -6$ i.e. $a|b$ and $b|a$ but $a \neq b$.

Hence R is not antisymmetric.

Equivalence Relation:

A relation R on a set A is called an equivalence relation if it is "reflexive, symmetric and transitive" - following three properties are fulfills:

(i) for every $a \in A$, aRa i.e. $(a,a) \in R$ (Reflexive)

(ii) If aRb then bRa i.e. $(a,b) \in R \Rightarrow (b,a) \in R$ (Symmetric)

(iii) If aRb and bRc then aRc i.e. $(a,b) \in R, (b,c) \in R \Rightarrow (a,c) \in R$

Q: If R be a relation in set of Integers defined by $R = \{(x,y) : x \in Z, y \in Z, (x-y)$ is divisible by $6\}$ then prove that R is an equivalence relation.

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Solution : (1) Reflexive. Let $x \in Z$, then
$x - x = 0$ and 0 is divisible by 6 (any no).
$\therefore x R x \ \forall x \in Z$. Hence it is reflexive

(2) Symmetric: Let $x, y \in Z$, then and xRy
i.e $(x-y)$ is divisible by 6
$\Rightarrow -(x-y)$ is also divisible by 6.
$\Rightarrow (y-x)$ is divisible by 6.
$\Rightarrow yRx$
So, R is symmetric.

(3) Transitive. $x, y, z \in Z$
and xRy, yRz
i.e $(x-y)$ is divisible by 6 and $(y-z)$ is divisible by 6
$\Rightarrow [(x-y)+(y-z)]$ is divisible by 6
$\Rightarrow (x-z)$ is divisible by 6.
$\Rightarrow xRz$.
So, R is transitive
Hence R is an equivalence relation.

Q Let N be the set of all natural nos. and let R be a relation on $N \times N$, defined by $(a,b) R (c,d) \Leftrightarrow ad = bc \ \forall (a,b), (c,d) \in N \times N$. Prove that R is an equivalence relation on $N \times N$.

Sol^n: (1) Reflexive: Let $(a,b) \in N \times N \Rightarrow a, b \in N \Rightarrow ab = ba$ is true
i.e $(a,b) R (a,b) \ \forall (a,b) \in N \times N \rightarrow$ Reflexive
(2) Symmetric: Let $(a,b), (c,d) \in N \times N$ such that $(a,b) R (c,d)$
then ~~ac=bd~~ $\Rightarrow$ ~~da=cb~~ $ad = bc \Rightarrow da = cb$.
$\Rightarrow cb = da$
$\Rightarrow (c,d) R (a,b) \ \forall (a,b), (c,d) \in N \times N$
So, R is symmetric.

(3) Transitive: Let $(a,b), (c,d), (e,f) \in N \times N$ such that $(a,b) R (c,d)$ and $(c,d) R (e,f)$. then
$\Rightarrow ad = bc$ & $cf = de \Rightarrow (ad)(cf) = (bc)(de)$

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Q: $N \to$ set of all natural nos. and R be the Relation on $N \times N$ defined by $(a,b) R (c,d) \Leftrightarrow ad(b+c) = bc(a+d)$ (5)

Check whether R is an equivalence Relation on $N \times N$.

Sol^n - Reflexive

Let $(a,b) \in N \times N \Rightarrow a, b \in N$

$\Rightarrow ab(b+a) = ba(a+b) \Rightarrow (a,b) R (a,b)$

$\therefore (a,b) R (a,b) \ \forall \ (a,b) \in N \times N \Rightarrow R$ is Reflexive

- Symmetric : Let $(a,b), (c,d) \in N \times N$.

and $(a,b) R (c,d)$

$\Rightarrow ad(b+c) = bc(a+d)$.

$\Rightarrow da(c+b) = cb(d+a)$.

$\Rightarrow cb(d+a) = da(c+b)$.

$\Rightarrow (c,d) R (a,b)$

i.e. $(a,b) R (c,d) \Rightarrow (c,d) R (a,b) \therefore$ R is symmetric.

- Transitive Let $(a,b), (c,d), (e,f) \in N \times N$, such that

$(a,b) R (c,d)$ and $(c,d) R (e,f)$, then

$\Rightarrow ad(b+c) = bc(a+d)$

$\Rightarrow \frac{b+c}{bc} = \frac{a+d}{ad} \Rightarrow \frac{1}{c} + \frac{1}{b} = \frac{1}{d} + \frac{1}{a}$. — (1)

and $cf(d+e) = de(c+f)$

$\Rightarrow \frac{d+e}{de} = \frac{c+f}{cf} \Rightarrow \frac{1}{e} + \frac{1}{d} = \frac{1}{f} + \frac{1}{c}$ — (2)

Adding (1) & (2), we get

$\left(\frac{1}{c} + \frac{1}{b}\right) + \left(\frac{1}{e} + \frac{1}{d}\right) = \left(\frac{1}{d} + \frac{1}{a}\right) + \left(\frac{1}{f} + \frac{1}{c}\right)$

$\Rightarrow \frac{1}{b} + \frac{1}{e} = \frac{1}{a} + \frac{1}{f}$

$\Rightarrow \frac{(e+b)}{be} = \frac{(f+a)}{af} \Rightarrow af(e+b) = be(f+a)$

$\Rightarrow af(b+e) = eb(a+f)$

$\Rightarrow (a,b) R (e,f)$

R is Transistive in $N \times N$

Hence R is an equivalence Relation on $N \times N$.

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Composite Relation: The composition relation of two relations

Let R be a relation from set A to B and S a relation from set B to C. The composite of R and S is the relation consisting of ordered pair (a, c) where $a \in A$ and $c \in C$, for which there exist an element $b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$. The composite relation denoted by $S \circ R$. ie

$$S \circ R = \{(a,c) : a \in A \text{ & } c \in C \text{ and } \exists \text{ an element } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S\}$$

Hence $(a,b) \in R$, and $(b,c) \in S \Rightarrow (a,c) \in S \circ R$.

Th: If A, B, C are three sets and $R \subseteq A \times B$, $S \subseteq B \times C$, then prove that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$

Proof: Let $(c,a) \in (S \circ R)^{-1} \Leftrightarrow (a,c) \in S \circ R$

$\Leftrightarrow \exists\, b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$

$\Leftrightarrow \exists\, b \in B$ such that $(b,a) \in R^{-1}$ and $(c,b) \in S^{-1}$

$\Leftrightarrow \exists\, b \in B$ such that $(c,b) \in S^{-1}$ and $(b,a) \in R^{-1}$

$\Leftrightarrow (c,a) \in R^{-1} \circ S^{-1}$

Hence $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$

Q. What is the composite of the relations R and S, where R is the relation from $A = \{1,2,3\}$ to $B = \{1,2,3,4\}$ with $R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$ and S is the relation from $\{1,2,3,4\}$ to $C = \{0,1,2\}$ with $S = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$?

Sol^n Here $R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$, & $S = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$

$S \circ R = \{(1,0), (1,1), (2,2), (2,1), (3,0), (3,1)\}$

Composing the parent Relation with itself:

Let R be the relation on the set of all people such that $(a,b) \in R$ if person a is a parent of person b. Then $(a,c) \in R \circ R$ iff there is a person b such that $(a,b) \in R$ and $(b,c) \in R$.

Let R be a relation on the set A. The powers R^n, $n = 1,2,3,\ldots$ are defined recursively by $R^1 = R$ and $R^{n+1} = R^n \circ R$.

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$\Rightarrow$ $(c,a) \in R \Rightarrow (a,c) \in R^{-1}$

Hence R^{-1} is transitive. Therefore R^{-1} is an equivalence relation.

Equivalence classes

If R is an equivalence relation on a set A and xRy then x and y are called equivalent with respect to R. The set of all elements of A that are equivalent to a given element x constitute the equivalence class of x and is denoted by $[x]_R$.

If only one relation is in the set then subscript R is deleted and the equivalence class is denoted by $[x]$.

The collection of all equivalence classes of elements of A under an equivalence relation R is denoted by

$$A/R = \{ [x] : x \in A \}.$$

It is known as the quotient set of A by R.

Properties of Equivalence classes.

Let A be non-empty set and R be an equivalence relation defined on A. Let a and $b \in A$ be arbitrary.

Then

(i) $a \in [a]$ (ii) $b \in [a] \Rightarrow [b] = [a]$

(iii) $[a] = [b] \Leftrightarrow (a,b) \in R$ (iv) Either $[a] = [b]$ or $[a] \cap [b] = \phi$.

Proof: R being an equivalence relation, then it is reflexive.

ie. aRa and $[a] = \{ x : x \in A \text{ and } xRa \}$

Whence $aRa \Rightarrow a \in [a]$.

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(ii) Let $b \in [a] \Rightarrow bRa$

Suppose $x \in [b]$, then $x \in [b] \Rightarrow xRb$

Again as R is transitive, $xRb, bRa \Rightarrow xRa$

$\Rightarrow x \in [a]$

$\Rightarrow [b] \subseteq [a]$ — (1)

Now let $y \in [a] \Rightarrow yRa$

As R is symmetric, $\therefore bRa \Rightarrow aRb$.

So, yRa and $aRb \Rightarrow yRb$. as R is transitive

Hence $y \in [b]$

$\Rightarrow [a] \subseteq [b]$ — (2)

from (1) & (2)

$[a] = [b]$ if $b \in [a]$

(iii) We assume that $[a] = [b]$

Since R is reflexive, we have aRa

Again $aRa \Rightarrow a \in [a] \Rightarrow a \in [b]$

$\Rightarrow aRb$.

Hence $[a] = [b] \Rightarrow aRb$ i.e. $(a,b) \in R$

Conversely: We assume that aRb.

Let $x \in [a]$ so that xRa.

But aRb, hence R being transitive,

$xRa, aRb \Rightarrow xRb$.

$\therefore x \in [b]$

$x \in [a] \Rightarrow x \in [b]$

$\Rightarrow [a] \subseteq [b]$. — (A)

Again, let $y \in [b]$ be arbitrary.

Then, $y \in [b] \Rightarrow yRb$.

But R being symmetric, $aRb \Rightarrow bRa$

$yRb, bRa \Rightarrow yRa$ (R is transitive).

$\Rightarrow y \in [a]$

$[b] \subseteq [a]$ — (B)

from (A) and (B)

$[a] = [b]$.

But $[a] = [b] \Rightarrow aRb$

and $aRb \Rightarrow [a] = [b]$

$\therefore [a] = [b] \Leftrightarrow aRb$.

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Q Let R_5 be the relation on the set Z of integers, defined by

$$x \equiv y \pmod 5$$

which reads "x is congruent to y modulo 5" and which means that the difference $x-y$ is divisible by 5. Prove that R_5 is an equivalence relation.

Sol": To prove R_5 is an equivalence relation, we have to prove that it is "reflexive, symmetric and transitive".

(a) Reflexive: for any integer n, we have $n-n=0$, and 0 is divisible by 5. Thus nR_5n, and R_5 is Reflexive.

(b) Symmetric: Let xR_5y, so that $x-y$ is divisible by 5. Its negative $-(x-y)$ is also divisible by 5, i.e. $(y-x)$ is also divisible by 5.

So, yR_5x. Thus R_5 is symmetric.

(c) Transitivity: Let us assume that lR_5m and mR_5n.

We know that $l-m = 5k$, $m-n=5j$. for some integers k and j.

$\Rightarrow (l-m)+(m-n) = l-n = 5(k+j)$.

i.e. divisible by 5.

So, lR_5m, $mR_5n \Rightarrow lR_5n$. Thus R_5 is transitive.

These are exactly five equivalence classes in the quotient set Z/R_5 as follows:

$A_0 = \{ \ldots -10, -5, 0, 5, 10 \ldots \} = [0]$

$A_1 = \{ \ldots -9, -4, +1, 6, 11, \ldots \} = [1]$

$A_2 = \{ \ldots -8, -3, 2, 7, 12, \ldots \} = [2]$

$A_3 = \{ \ldots -7, -2, 3, 8, 13, \ldots \} = [3]$

$A_4 = \{ \ldots -6, -1, 4, 9, 14, \ldots \} = [4]$

$Z/R = \{ [0], [1], [2], [3], [4] \}$.

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Boolean Algebra

George Boole - 1854 : Investigation of the Laws of thoughts. → Values variables may assumes are 0 & 1. (False & True)

For any given algebra system, there are some initial assumptions (Postulates) that system follows.

Closure; Commutativity, Associativity, Distribution, Identity, Inverse.

(·) → logical AND, (+), logical OR, (') (Complement) logical NOT.

(x·y)	(x+y)	(x')
(x∧y)	(x∨y)	(¬x)
		(~x)

Def": Let A be a set, a, b, c, . . . are elements of A. Let ∪(+, ∨) and ∩(·, ∧) are binary operation on A, then the algebraic structure (A, ∪, ∩) is called a Boolean algebra, if the elements of A obey the following laws:

A1 Closure: for any a, b ∈ A,

(i) a ∪ b ∈ A (ii) a ∩ b ∈ A ∀ a, b ∈ A

A2: Commutative: for any a, b ∈ A

(i) a ∪ b = b ∪ a (ii) a ∩ b = b ∩ a ∀ a, b ∈ A.

A3: Associative: for any a, b, c ∈ A, then

(i) (a ∪ b) ∪ c = a ∪ (b ∪ c) (ii) (a ∩ b) ∩ c = a ∩ (b ∩ c) ∀ a, b, c ∈ A.

A4: Distributive: for any a, b, c ∈ A, then

(i) a ∪ (b ∩ c) = (a ∪ b) ∩ (a ∪ c) ∀ a, b, c, ∈ A.

(ii) a ∩ (b ∪ c) = (a ∩ b) ∪ (a ∩ c)

A5: Identity: There exists two elements 0 and 1 in A such that for every a ∈ A, we have

(i) a + 0 = 0 + a = a

(ii) a . 1 = 1 . a = a

A6: Laws of Inverse: (Complement laws) for each a ∈ A, ∃ a' ∈ A such that

(i) a + a' = a' + a = 1 (ii) a . a' = a' . a = 0

A7: 0 ≠ 1

on B defined as:

+	0	1
0	0	1
1	1	1

·	0	1	a	a'
0	0	0	0	1
1	0	1	1	0

Soln: From the given table

(1) Closure: It is clear that, both the operations are closed.

(2) Commutative:

$0+1=1=1+0$ & $0.1=1.0=0$

(3) Associative

$0+(1+0)=(0+1)+0$

and $0.(1.0)=(0.1).0$

(4) Distributive:

$0+(1.0)=(0+1).(0+0)$

$0=0+0=1.0=0$

and $1.(1+0)=(1.1)+(1.0)$

$=1+0$

$=1$

(5) Identity.

$0+0=0, \quad 0+1=1+0=1$

$0.1=1.0=0, \quad 1.1=1$

(6) Complementation. $a\in B \Rightarrow a'\in B$

$0'=1, \quad 1'=0$

(7) Clearly $0\neq 1$.

Hence $(B,+,\cdot)$ is a Boolean algebra.

Q2: Let B be the set of all positive divisors of 30 and the operations $\vee$ and $\wedge$ on B are defined as follows:

$a\vee b=$ LCM of a & b

$a\wedge b=$ HCF of a & b.

Prove that $(B,\vee,\wedge)$ is a Boolean algebra.

Sol: Here $B=\{1,2,3,5,6,10,15,30\}$

Composition ($\vee$)

$\vee$	1	2	3	5	6	10	15	30
1	1	2	3	5	6	10	15	30
2	2	2	6	10	6	10	30	30
3	3	6	3	15	6	30	15	30
5	5	10	15	15	30	10	15	30
6	6	6	6	30	6	30	30	30
10	10	10	30	10	30	10	30	30
15	15	30	15	15	30	30	15	30
30	30	30	30	30	30	30	30	30

$\wedge$ composition

$\wedge$	1	2	3	5	6	10	15	30
1	1	1	1	1	1	1	1	1
2	1	2	1	2	1	2	1	2
3	1	1	3	1	3	1	3	3
5	1	1	1	5	1	5	5	5
6	1	2	3	1	6	2	3	6
10	1	2	1	5	2	10	5	10
15	1	1	3	5	3	5	15	15
30	1	2	3	5	6	10	15	30

Complement

a	a'
1	30
2	15
3	10
5	6
6	5
10	3
15	2
30	1

1- Closure: From composition table, it is clear that $\vee$ and $\wedge$ are closed.

2. Commutative:
$\forall a, b \in B$, we have $a \vee b = b \vee a$
& $a \wedge b = b \wedge a$

For example $3 \vee 5 = 15 = 5 \vee 3$.

and $3 \wedge 5 = 1 = 5 \wedge 3$

3. Associative: for $a, b, c \in B$, we have
$a \vee (b \vee c) = (a \vee b) \vee c$ for $a, b, c \in B$
and $a \wedge (b \wedge c) = (a \wedge b) \wedge c$ for $a, b, c \in B$.

for Example $3 \vee (5 \vee 6) = 3 \vee 30 = 30$
$(3 \vee 5) \vee 6 = 15 \vee 6 = 30$
$\Rightarrow 3 \vee (5 \vee 6) = (3 \vee 5) \vee$

Also $2 \wedge (3 \wedge 10) = 2 \wedge 1 = 1$
$(2 \wedge 3) \wedge 10 = 1 \wedge 10 = 1$

$\Rightarrow 2 \wedge (3 \wedge 10) = (2 \wedge 3) \wedge 10$
So, $\vee$ & $\wedge$ associative

4. Distributive, For $a, b, c \in B$, we have
$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
and $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

for Example $2 \vee (3 \wedge 5) = 2 \vee 1 = 2$
$(2 \vee 3) \wedge (2 \vee 5) = 6 \wedge 10 = 2$
$\Rightarrow 2 \vee (3 \wedge 5) = (2 \vee 3) \wedge (2 \vee 5)$

Similarly $2 \wedge (3 \vee 5) = (2 \wedge 3) \vee (2 \wedge 5)$
$\Rightarrow 2 \wedge (15) = 1 \vee 2 \Rightarrow 2 = 2$

5: Identity Element. Identity element for $\vee$ is 1 and identity element for $\wedge$ is 30, since $\forall a \in B$, we have:

$$a \vee 1 = 1 \vee a = a$$
$$\& \quad a \wedge 30 = 30 \wedge a = a.$$

Eg: $15 \vee 1 = 1 \vee 15 = 15$

$5 \wedge 30 = 30 \wedge 5 = 5$

6: Complementation: for each $a \in B$, $\exists$ its complement element such that

$$a \vee b = 30$$
$$\& \quad a \wedge b = 1$$

Eg: $3 \vee 10 = 30$.

and $3 \wedge 10 = 1$

Hence $(B, \vee, \wedge)$ is a Boolean Algebra.

Properties of Boolean Algebra.

(I) Idempotent law: for every element of Boolean algebra B, (i) $a+a=a$ (ii) $a.a=a$, $\forall a \in B$

(i) LHS $= a+a = (a+a).1$ (Identity Element).

$= (a+a).(a+a')$ (Complement)

~~$= (a.1)+(a.1)$~~ (Distributive)

$= (a.a) + (a.a')$ (Distributive)

$= a + 0$ (complement)

$= a$ (Identity)

$=$ RHS

(ii) TP $a.a = a$

LHS $= a.a$

$= a.a + 0$ (Identity element)

$= a.a + a.a'$ (Complement.

$= a.(a+a')$ (Distributive).

$= a.1$. (~~Complement~~ Identity)

$= a$ Complement.

(II) for every element $a \in B$,

(i) $a+1=1$ (ii) $a.0=0$

(i) TP $a+1=1$, LHS $= a+1$ → Identity
$= a+(a+a')$ → Assoc
$= (a+a)+a'$ → Idempotent
$= a+a'$ (Identity)
$= 1$

(ii) TP $a.0 = 0$
LHS $= a.0 = a.(a.a')$ Identity
$= (a.a).(a')$ Assoc
$= (a.a')$ Idempotent
$= 0$ (Identity)

(IV) Absorption laws:
For any $a, b \in B$,
(i) $a+(a.b)=a$, (ii) $a.(a+b)=a$

Pf (i) TP $a+(a.b)=a$
LHS $= a+(a.b) = a.1 + a.b$ (Identity)
$= a.(1+b)$ (Distributive)
$= a.1$ $(1+b=1$, proved$)$
$= a$ (Identity)

(ii) TP $a.(a+b)=a$
LHS $= a.(a+b)$ $= (a+0).(a+b)$ (Identity)
$= a+(0.b)$ $= a+0 = a$
Dis. $(a.0=a)$

(V) Associative
(i) $a+(b+c) = (a+b)+c$
(ii) $a(bc) = (ab)c$

Pf: (ii) first we take $a+a(bc)=a+(ab)c$ — (1)
LHS $= a+a(bc)$
$= a$ (absorption)
$= a(a+c)$ (absorption)
$= (a+ab)(a+c)$ (absorption)
$= a+(ab)c$ (Distribut & Identity)
RHS of (1)

$(a+b)(a+c) = (a+b)(a.c)+(bc)$

Again $a'+a(bc) = a'+(ab)c$ — (2)
LHS $= a'+a(bc)$
$= (a'+a)(a'+bc)$ (Dist)

$= [(a+a')(a'+b)](a'+c)$

$= [a'+ab](a'+c)$ (Dist).

$= a'+(ab)c$ = RHS (2).

By Multiplying (1) & (2).

$[a+a(bc)][a'+a(bc)] = [a+(ab)c][a'+(ab)c]$ ③

LHS $= (a+a(bc))(a'+a(bc))$

$= (a(bc)+a)(a'+a(bc))$ - Commutat.

$= a(bc)+aa'$ (Dist)

$= a(bc)+0$ - (Identity)

$= a(bc)$ - ④

Similarly RHS $= [a+(ab)c][a'+(ab)c]$

$= (ab)c$ - ⑧

From ④ & ⑧ we get

$\boxed{a(bc) = (ab)c}$

Principle of Duality :

Let $(B, +, \cdot)$ be a Boolean ~~algebra~~ algebra, if in the statement of $(B, +, \cdot)$, the positions of '+' and '.' are interchanged and also the positions of Identity elements 0 and 1 are interchanged, the new statement is called the dual of the original statement.

Eg : $(1+a).(b+0) = b$

⇒ Its dual is $(0+a)+(b\cdot 1) = b$.

By dual property,

$a+(b+c) = (a+b)+c$

(v) In a Boolean algebra B, of $0,1 \in B$, then
(i) $0' = 1$ (ii) $1' = 0$.

(vi) The identity elements in B are unique.

(vii) The complement of each element of B.A. is unique.

Pf: Let $a \in B$ be any element. Suppose x & y are two complements of a. Then by Def. we have $a + x = x + a = 1$ and $a \cdot x = x \cdot a = 0$ } -1.
$a + y = y + a = 1$ and $a \cdot y = y \cdot a = 0$ }

Now $x = x \cdot 1$ (Identity)
$= x \cdot (a + y) = x \cdot a + ay$ (Dist).
$= 0 + xy$ (Identity & Compl)
$= ay + xy$
$= (a + x) \cdot y$ → Compl.
$= 1 \cdot y$
$= y$

(viii) Involution law $(a')' = a$.

(ix) De Morgan's Law
(i) $(a+b)' = a' \cdot b'$, (ii) $(a \cdot b)' = a' + b'$

Pf: To prove The complement of $(a+b)$ is $a' \cdot b'$.
It is sufficient to prove that
$(a+b) + a' \cdot b' = 1$
and $(a+b) \cdot (a' \cdot b') = 0$ }.

Now $(a+b) + (a' \cdot b') = ((a+b) + a') \cdot ((a+b) + b')$
$= ((a + a') + b) \cdot (a + (b + b'))$ (Dist)
$= (1 + b) \cdot (a + 1)$ (Com & Compl)
$= 1 \cdot 1$ (Identity).
$= 1$.

and $(a+b) \cdot (a' \cdot b') = (a' \cdot b') \cdot (a + b)$ (Commutative)
$= (a' \cdot b') a + (a' \cdot b') b$
$= b'(a'a) + a'(ba)$
$= b' \cdot 0 + a' \cdot 0$

Similarly $(a.b)' = a' + b'$

Subboolean Algebra

A non-empty subset B_1 of B is said to be a Subboolean algebra of B if B_1 contains 0 and 1 together with two binary operations $(+)$ and $(.)$ and is closed under the above operations and complementation and is itself a Boolean Algebra.

The necessary and suff. condn. for sub boolean algebra is that (i) $a, b \in B_1 \Rightarrow (a.b) \in B_1$

(ii) $a \in B_1 \Rightarrow a' \in B_1$.

Q(1)

In a B. Alg. B prove that:

(i) $[x'(x+y)]' + [y(y+x')]' + [y(y'+x)]' + [y(y'+x)]' = 1$.

LHS $= [x'(x+y)]' + [y(y+x')]' + [y(y'+x)]'$

$= (x')' + (x+y)' + y' + (y+x')' + y' + (y'+x)'$ (De Morgans)

$= x + x'y' + y' + y'x + y' + yx'$ (De Morgan)

$= x + xy' + y' + y' + x'y' + yx'$

$= x(1+y') + y' + x'y' + yx'$

$= x.1 + y'(1+x') + yx'$

$= x + y' + yx'$

$= x + yx' + y'$

$= (x+x')(x+y) + y'$

Dist.

$= 1(x+y) + y' = (x+y) + y' = x + (y+y')$

$= x + 1$

(ii) $(x+yz)(y'+x)(y'+z)' = x(y'+z')$ $= 1$

LHS $= (x+y)(x+z)(x+y')(y'+z')$

$= (x+y)(x+y')(x+z)(y'+z')$

$= (x+yy')(x+z)(y'+z')$

$= x.(x+z)(y'+z') = (x+xz)(y'+z')$

$= x(1+z)(y'+z') = x.1(y'+z')$

$(a+a'b = a)$

$= x(y'+z')$ RHS.

Boolean Function

Constant 0, 1.

→ Variable 'x' is called a Boolean variable if it assumes values either 0 or 1.

Boolean expressions are formed by application of the basic operations to one or more variables or constants.

" Any expression that contains 'n' distinct variables is usually referred to as a Boolean expression of n variables. Each appearence of a variable or its complement in an expression is referred as literal.

" Any exp. which is the combination of a finite set of symbols, combination of constants & variables by the operation of '+' and '.' is called a Boolean fⁿ.

⇒ Thus an Boolean Exp. is a boolean fⁿ.

$f_0 = AB+C.$

Algebraic Manipulation of Boolean Expression

Ex. (1) $ab + ab' + a'b = a(b+b') + a'b$

$= a.1 + a'b$

$= a + a'b = (a+a')(a+b)$

$= (a+b)$

(2) $(a'b + a'b' + b')' = (a'(b+b') + b')'$

$= (a'.1 + b')' = (a'+b') = \underline{ab}$

(3) $b(a+c) + ab' + bc' + c = ba + bc + ab' + bc' + c.$

$= ab + ab' + bc + bc' + c$

$= a(b+b')$

Sum of products and product of sums form:

A Boolean expression E is said to be in a sum of products form if E is a sum of two or more products of variables.

Ex $x_1x_2x_3 + x_1'x_3 + x_1x_2'x_3$;

Soln $a'b + abc' + abc$, $xz' + x'yz + xy'z + xy$.

One complementation sign can not cover more than one variable in a term.

Product of Sum form: (i) $(x+y+z)(x'+y+z')$

(ii) $(x_1+x_2'+x_3)(x_1+x_3)$

Fundamental Forms of Boolean Functions

Minimal Boolean function:

In n variables, the product of the 'n' variables taken together is called minimal Boolean function of 'n' variables.

Ex For two variables: $x \& y$: xy, $x'y$, xy', $x'y'$.

For three: (x,y,z) : xyz, xyz', $xy'z$, $x'yz$, $xy'z'$, $x'yz'$, $x'y'z$, $x'y'z'$.

Thus, minimal function of 'n' variables will consist 2^n elements.

Boole's Theorem:

It states that there are 2^n minimal Boolean functions in 2^n elements in 'n' variables.

Normal Form of Boolean Function:

A Boolean function of 'n' variables is said to be in normal form if it consists of all n variables.

Ex: $x+y+z$, $x'+y+z'$, $x+x'y+yz$ etc.

Canonical forms:

With the help of 'n' input variables, an infinite number of possible logic expressions can be constructed. There are infinite no. of logic expressions that are equivalent. To help eliminate possible confusion, logic designers have developed standard forms for logic expressions.

~~A unique can~~
using a canonical or standardized form

Sum of Minterms → It is referred to as disjuctive normal form.

Product of Maxterm: ——— to a conjuctive normal

Disjunctive Normal Form (DNF) (Sum of product) form.

Variable	DNF
1 Variables	$x+x'$
2 Variables	$xy+x'y+xy'+x'y'$
3 Variables	$xy'z+x'yz+xyz+x'y'z'$ etc.

Q1 $(x+y+z)(xy+x'z)'$, to DNF

$= (x+y+z)(xy+x'z)' = (x+y+z)[(xy)'\cdot(x'z)']$

$= (x+y+z)[(x'+y')][(x+z')]$

$= (x+y+z)(x'z'+x'y+xy'+y'z')$

$= (x+y+z)(x'z'+xy'+z'x)$

$= x'z'+xy'+z'x+x'xz'+xy'y+z'yx$

$= x'z'+xy'+z'x+xy'+z'yx$

$= x'z'+xy'+z'x+z'yx$

$= xy'+x'z'+z'x+z'yx$

$= xy'+x'z'+z'x(1+y)$

$= xy'+x'z'+z'x$

Q1: $f(x,y,z) = z'y+x$ (DNF)

$= z'y+(y+y')x$

$= z'y+xy+y'x$

$= z'y+xy+y'x+z'y(x+x')$

$= z'y+xy+y'x+z'yx+z'yx'$

$= z'y+xy+y'x+z'yx+z'yx'$

$= xy+y'x+z'yx+z'yx'$

$= xy+z'yx+y'x+z'yx'$

$= xy+z'yx+x(y'+z'y)$

$= xy+z'yx+x(y'+z')(y'+y)$

$= xy+z'yx+x(y'z'+y'+z'y)$

$= xy+z'yx+xy'z'+xy'+xz'y$

$= xy+xy'+z'yx+xy'z'+xz'y$

$= x(y+y')+z'yx+xy'z'+xz'y$

$= x+z'yx+xy'z'+xz'y$

$= x(1+z'y+y'z'+z'y)$

$= x$ (DNF)

Q CNF (i) $(x+y+z)(xy+x'z)'$

$= (x+y+z)[(xy)' \cdot (x'z)']$

$= (x+y+z)(x'+y')(x+z')$

$= (x+y+z)(x'+y'+zz')(x+yy'+z')$

$= (x+y+z)(x'+y'+z)(x'+y'+z')(x+z'+y)(x+y'+z')$

(ii) $xy' + xz + yz$

$= x(y+y') + xz = x + xz = x(1+z) = x \cdot 1 = x$

$= x+0 = (x+yy') = (x+y)(x+y')$

$= (x+y+zz')(x+y'+zz')$

$= (x+y+z)(x+y+z')(x+y'+z)(x+y'+z')$

Conversion of Disjunction Normal form to its CNF and vice-versa

DNF to CNF

(i) Let DNF of (ii) find f' which is the sum of those terms of the complete DNF not present in (iii) Take complement of f' i.e. f'.

(iv) By DeMorgan's —

CNF to DNF

1 — Call it f

2 — find f' by not present in.

3 — Take f' of f'

4 — Simply by DeMorgan's —

Q CNF to DNF

$f = (x'+y+z')(x'+y+z)(x+y'+z)$

$f' = [(x+y+z)(x'+y'+z')(x+y'+z')(x'+y'+z)(x+y+z')]$

$f = [\quad]'$

$= [(x+y+z)(x'+y'+z')(x+y'+z')(x'+y'+z)]' + (x+y+z')'$

$= [(x+y+z)(x'+y'+z')(x+y'+z')]' + (x'+y'+z) + (x'+y+z)$

$= [(x+y+z)(x'+y'+z')]' + (x+y'+z')' + (x'y+z) + (x'y+z)$

$= (x+y+z)' + (x'+y'+z')' + (x'y+z) + (x'y+z)$

Q change from DNF to CNF in three variables.

(i) $uv + u'v + u'v'$

$= (u+u')v + u'v'$

$= 1 \cdot v + u'v' = v + u'v'$

$= (v+u')(v+u') = (v+u') \cdot 1 = (v+u')$

$= (v+u') + ww' = (v+u'+w) \cdot (v+u'+w')$.

(ii) If $f(x,y,z) = xyz + xy'z + x'yz' + x'y'z + x'y'z'$

$f'(x,y,z) = xy'z' + xyz' + x'yz$

Now $[f'(x,y,z)]' = [(xy'z') + (xyz') + (x'yz)]'$

$= [(xy'z') + (xyz')]' \cdot (x+y+z')$

$= (x'+y+z) + (x'+y'+z) + (x+y+z')$

$= (xy'z')' \cdot (xyz')' \cdot (x'yz)'$

$= (x'+y+z)(x'+y'+z)(x+y+z')$

Prime Implicals

A fundamental product P is called a prime implicant of a Boolean expression E, if

$$P + E = E$$

Ex Suppose $E = xy' + xyz' + x'yz'$

we can check that

$$xz' + E = E$$

i.e. $xz' + xy' + xyz' + x'yz' = xy' + xy' + x'yz'$

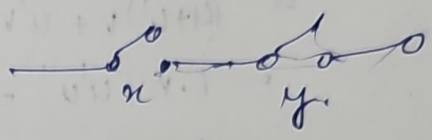
Point $x + E \neq E$ and $z' + E \neq E$ (Distribution)

So, xz' is a prime ~~implicant~~ implicant of E

Switching Circuits

ON	OFF
~~Ope~~ Closed	Open
1	0

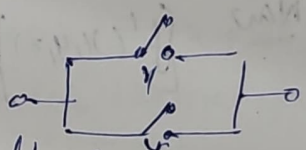
In Series



x	y	x.y
0	0	0
0	1	0
1	0	0
1	1	1

In Parallel

x	y	x+y
0	0	0
0	1	1
1	0	1
1	1	1



Properties of S.C.

(1) Commutative $x+y = y+x$

$x \cdot y = y \cdot x$

(2) Distributive.

$x \cdot (y+z) = (x \cdot y) + (x \cdot z)$

$x + (y \cdot z) = (x+y) \cdot (x+z)$

(3) Identity $x+0 = x$

$x \cdot 1 = x$

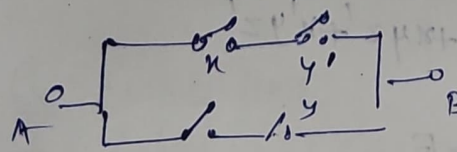
(4) Complement $x + x' = 1$

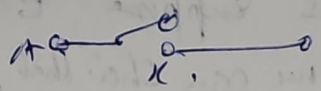
$x \cdot x' = 0$.

Q Draw the simplified switching circuit for

$f(x,y) = x \cdot y' + x \cdot y = x(y'+y)$

$= x \cdot 1 = x$





Q Simplify the circuit

$f(x,y) = [(x+y)(x+y') \cdot x] + yy']$

$= [x(x+y') + y(x+y')]x + 0$

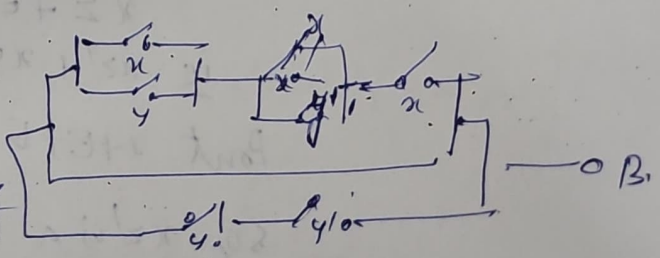
$= [xx + xy' + yx + yy']x + 0$

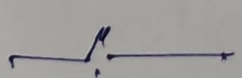
$= [x + xy' + yx + 0]x + 0$.

$= x \cdot x + x \cdot xy' + yxx$

$= x + xy' + yx = x(1+y) + xy' = x + xy' = x(1+y')$

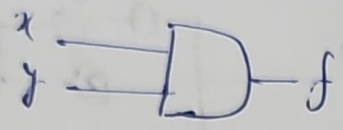
$= x$





Logic Circuits:

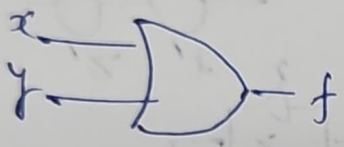
AND



x	y	f
0	0	0
0	1	0
1	0	0
1	1	1

$f = xy$

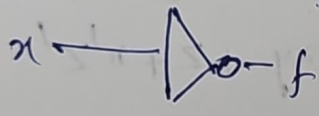
OR



x	y	f
0	0	0
0	1	1
1	0	1
1	1	1

$f = x + y$

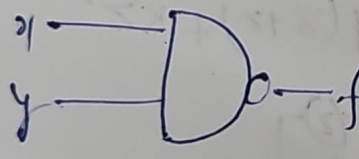
NOT (Inverter)



x	f
0	1
1	0

$f = x'$

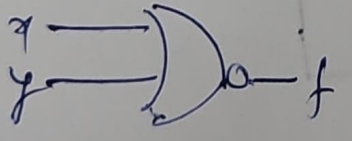
NAND



x	y	f
0	0	1
1	0	1
0	1	1
1	1	0

$f = (xy)'$

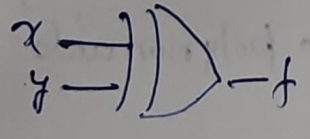
NOR



x	y	f
0	0	1
1	0	0
0	1	0
1	1	0

$f = (x+y)'$

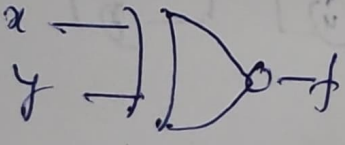
XOR (Exclusive-OR)



x	y	f
0	0	0
0	1	1
1	0	1
1	1	0

$f = x \oplus y$
$= x'y + xy'$

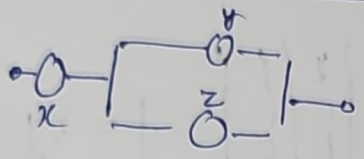
XNOR (Exclusive-NOR)

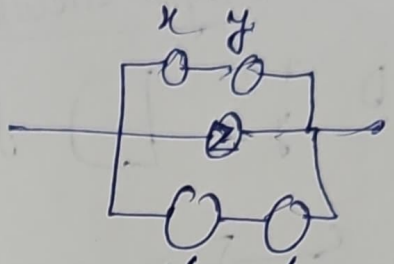


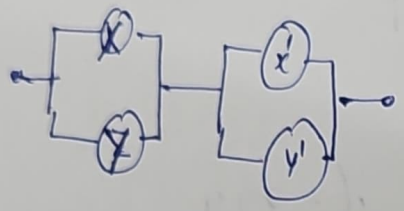
x	y	f
0	0	1
0	1	0
1	0	0
1	1	1

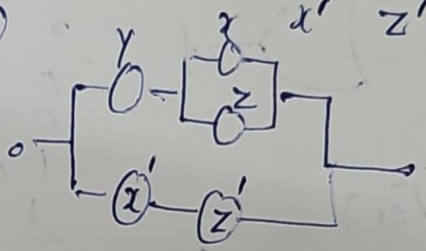
$f = (x \oplus y)'$
$= (xy + x'y')$

Q switches functions:

(1) 

(2) 

(3) 

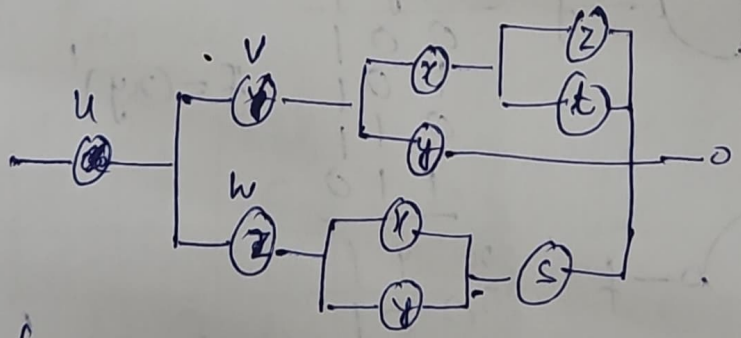
(4) 

(1) $f = x(y+z)$

(2) $f = xy + z + x'z'$

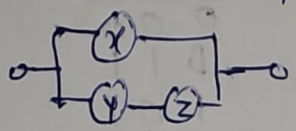
(3) $f = (x+y)(x'+y')$

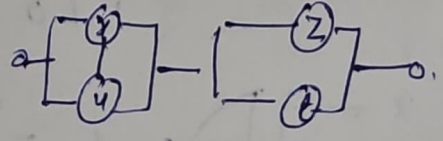
(4) $f = y(x+z) + x'z'$

(5) 

$$f = u\left[v\left[x(z+t)+y\right] + sw\left[x+y\right]\right]$$

Q Design the circuit for the following polynomials

(i) $x + yz$ 

(ii) $(x+y)(z+t)$ 

(iii) $x + y(z+st) + uv$

